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FACULTY OF ENGINEERING
TECHNOLOGY

CAMPUS GROUP T

Completion document for Lab Electricity

Circuit theorems

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1 SUPERPOSITION THEOREM

1.1 Circuit diagram with measured values

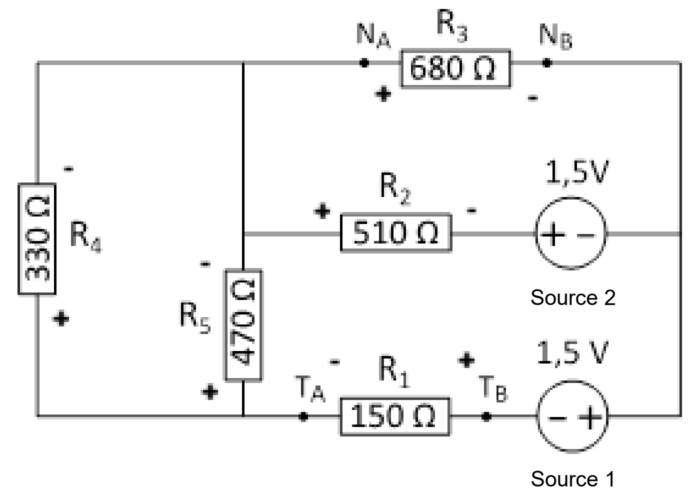

Figure 1: Circuit diagram

1.2 Measurement results

R	$R_{nominal}$ (Ω)	$R_{measured}$ (Ω)	$V_{measured\ total}$ (V)	V_1 (V)	V_2 (V)	V_{12} (V)
R1	150	148.5	-0.555	-0.202	-0.352	-0.554
R2	510	505	-1.73	-1.039	-0.69	-1.729
R3	680	679	-0.223	0.465	-0.69	-0.225
R4	330	327.1	-0.719	-0.262	-0.457	-0.719
R5	470	466	-0.719	-0.262	-0.457	-0.719

1.3 Calculations

According to the superposition theorem, the total voltage across a resistor should be equal to the algebraic sum of the voltages produced by each source acting independently:

$$V_{total} = V_1 + V_2$$

This relation is verified for each resistor. For example, for resistor R1:

$$V_{12} = (-0.202) + (-0.352) = -0.554 \text{ V}$$

This value closely matches the measured total voltage:

$$V_{measured\ total} = -0.555 \text{ V}$$

The resistor with the smallest total current is R3, since the voltage across it is relatively small compared to its resistance value. Therefore, the relative measurement error is expected to be most significant for this resistor.

For R3:

$$V_{measured\ total} = -0.223 \text{ V}$$
$$V_{12} = 0.465 + (-0.690) = -0.225 \text{ V}$$

The relative error is calculated as:

$$\text{Relative error} = \frac{|\ V_{measured\ total} - V_{12}\ |}{|\ V_{measured\ total}\ |} \times 100\%$$
$$= \frac{|\ -0.223 - (-0.225)\ |}{0.223} \times 100\%$$
$$= \frac{0.002}{0.223} \times 100\% \approx 0.90\%$$

1.4 Discussion

The measured total voltages across all resistors correspond closely to the algebraic sum of the partial voltages obtained when each source is considered separately. For each resistor, the relation $V_{total} \approx V_1 + V_2$ is satisfied with only very small deviations. These deviations can be explained by experimental uncertainties, such as the limited precision of the voltmeter, small variations in the measured resistance values, and contact resistances in the circuit. The relative error is largest for resistor R3, which carries the smallest current, making the measurement more sensitive to such inaccuracies. Based on these observations, it can be concluded that the superposition theorem is experimentally verified within the accuracy of the measurements.

2 THÉVENIN'S THEOREM

· Measured Thévenin resistance:

$$R_{Th,m} = 482\ \Omega$$

· Measured Thévenin voltage:

$$V_{Th,m} = -2.364\ \mathrm{V}$$

Figure 2 shows the circuit diagram used to determine the Thévenin resistance. In this setup, all independent power sources are turned off and replaced by short circuits in order to measure the equivalent resistance seen from terminals A and B.

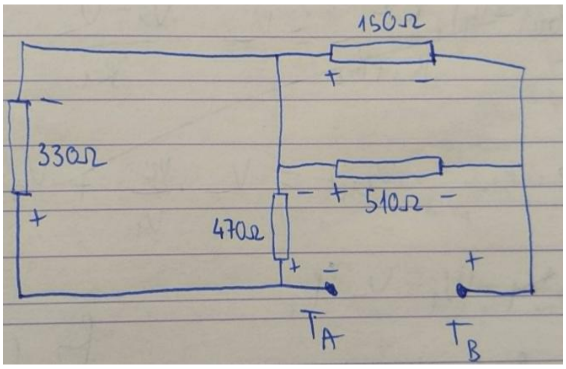

Figure 2: Circuit diagram without power sources

· Realized Thévenin equivalent:

The realized Thévenin equivalent circuit was constructed using a DC power supply and a resistor adjusted to the measured Thévenin resistance value. The setup used to experimentally verify Thévenin's theorem is shown in Figure 3.

Thévenin resistance:

$$R_{Th,r} = 482\ \Omega$$

Thévenin voltage:

$$V_{Th,r} = -2.360\ \mathrm{V}$$

· Measured voltage across R_1:

$$V_{R1,r} = -0.554\ \mathrm{V}$$

From Part 1, the measured voltage across R_1 in the original circuit is:

$$V_{R1,part1} = -0.555 \text{ V}$$

Comparison:

$$-0.554 \text{ V} \approx -0.555 \text{ V}$$

The difference is:

$$| -0.554 - (-0.555) | = 0.001 \text{ V}$$

which is negligible.

The voltage across R_1 in the realised Thévenin equivalent circuit is practically identical to the voltage measured in the original circuit. This shows that the original network can be replaced by its Thévenin equivalent without affecting the behavior of the load resistor. Therefore, Thévenin's theorem is experimentally verified.

· Calculation of theoretical expected Thévenin equivalent:

To determine the theoretically expected Thévenin equivalent, the measured resistance values and source voltages are used.

The Thévenin voltage V_{Th} is defined as the open-circuit voltage between terminals A and B with R_1 removed from the circuit. This voltage can be determined using Kirchhoff's laws or by applying the superposition theorem and summing the contribution of each source algebraically.

Figure 3 shows the realised Thévenin equivalent circuit used to verify the theorem experimentally.

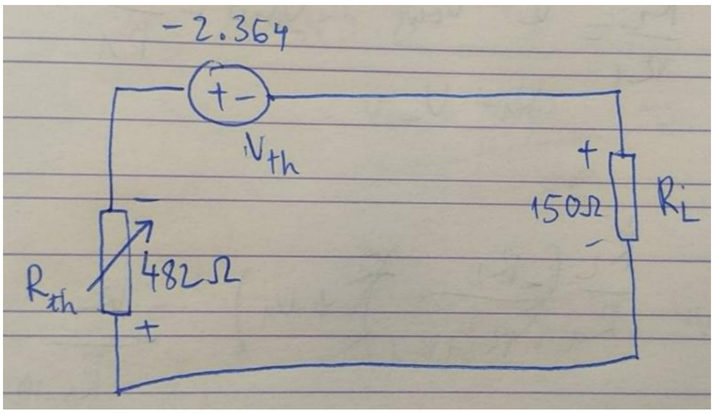

Figure 3: Thévenin equivalent circuit

Using the measured source voltages and resistor values, the theoretical Thévenin voltage is found to be:

$$V_{Th,theo} \approx -2.36\ \text{V}$$

The Thévenin resistance R_{Th} is determined by turning off all independent voltage sources and calculating the equivalent resistance seen from terminals A and B.

Using the measured resistor values, the equivalent resistance is calculated as:

$$R_{Th,theo} \approx 482\ \Omega$$

The theoretical values are in very good agreement with the measured results. Small differences can be explained by measurement uncertainties, contact resistances, and rounding effects.

3 NORTON'S THEOREM

· Measured value of Norton resistance:

$$R_{N,m} = 482\ \Omega$$

· Measured "short circuit" current:

$$I_{N,m} = -976\ \mu A$$

· Measured ammeter voltage:

$$V_A = 0.098\ \text{V}$$

· Corrected Norton current (with calculation):

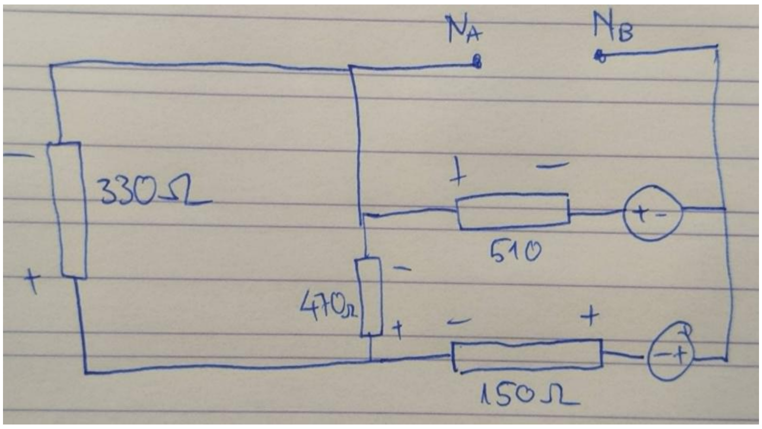

Figure 4: Circuit diagram for Norton experiment

In the Norton equivalent circuit, the ideal current source $I_N$is connected in parallel with the Norton resistance R_N. During the measurement of the short-circuit current, the ammeter is connected across the output terminals. Since the ammeter has a non-zero internal resistance, a voltage appears across it and part of the current flows through the Norton resistance instead of through the ammeter.

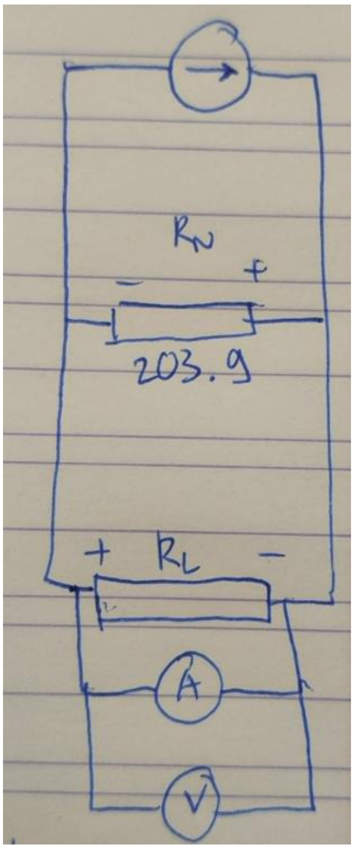

Figure 5: Norton equivalent circuit

$\mathrm{R_L}$ is removed when measuring current with ammeter.

The current flowing through the Norton resistance is:

$$I_{R_N} = \frac{V_A}{R_N}$$

Applying Kirchhoff's current law:

$$I_N = I_{meas} + I_{R_N}$$

Thus:

$$I_N = I_{meas} + \frac{V_A}{R_N}$$

Substituting the measured values:

$$I_N = -976 \times 10^{-6} + \frac{-0.098}{482}$$
$$I_N = -976 \times 10^{-6} - 203 \times 10^{-6}$$
$$I_N \approx -1179\ \mu A$$

The corrected Norton current has a greater magnitude than the measured current because the internal resistance of the ammeter causes part of the current to bypass the meter during the measurement.

4 OPTIONAL: PERSONAL REMARKS